The documentation and process conversion measures necessary to comply with this revision shall be completed by 12 August 2003.

INCH-POUND

MIL-PRF-19500/382D
12 May 2003
SUPERSEDING
MIL-PRF-19500/382C
28 July 1999

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPE 2N2944A, 2N2945A, 2N2945AM AND 2N2946A, 2N2944AUB, 2N2945AUB AND 2N2946AUB
JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for low-power, high-speed chopper, PNP, silicon transistors. A 'M' suffix will indicate a matched pair. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for each unencapsulated device type.

* 1.2 Physical dimension. See figure 1 (similar to T0-46), figure 2 (surface mount), and figure 3 (die).

* 1.3 Maximum ratings.

Types	P_T T_A = +25°C (1) (2)	V_{EBO}	V_{CBO}	V_{CEO}	V_{ECO}	I_C	T_J and T_{STG}	$R_{\theta JA}$ (3)
	mW	V dc	V dc	V dc	V dc	mA dc	°C	°C/W
2N2944A	400	-15	-15	-10	-10	-100	-65 to +200	435
2N2945A	400	-25	-25	-20	-20	-100	-65 to +200	435
2N2946A	400	-40	-40	-35	-35	-100	-65 to +200	435
2N2944AUB	400	-15	-15	-10	-10	-100	-65 to +200	435
2N2945AUB	400	-25	-25	-20	-20	-100	-65 to +200	435
2N2946AUB	400	-40	-40	-35	-35	-100	-65 to +200	435

(1) For derating, see figures 4, 5 and 6.
(2) See 3.3.
(3) For thermal curves, see figures 7.
(4) For non-thermal conductive PCB or unknown PCB surface mount conditions in free air, substitute figures 7 and 14 for the UB package and use $R_{\theta JA}$.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAC, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics.

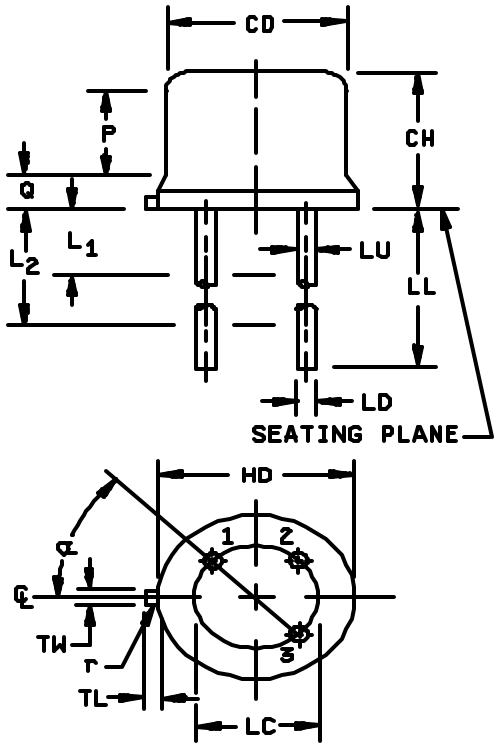
Limits	h_{FE1} V_{CE} = -0.5 V dc I_C = -1 mA dc			h_{FE} (inv)$_1$ V_{EC} = -0.5 V dc I_E = -200 µA dc			r_{ec} (on)$_2$ I_B = -1 mA dc I_e = 100 µA ac(rms) I_E = 0; f = 1 kHz		
	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A
							ohms	ohms	ohms
Min	100	70	50	50	30	20			
Max							4	6	8

Limits	V_{EC} (ofs)						$\lvert h_{fe} \rvert$		
	I_B = -200 µA dc I_E = 0			I_R = -1.0 mA dc I_E = 0			f = 1 MHz V_{CE} = -6 V dc I_C = -1 mA dc		
	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A
	mV dc	mV dc	mV dc	mV dc	mV dc	mV dc			
Min							15	10	5
Max	-0.3	-0.5	-0.8	-0.6	-1.0	-2.0	55	55	55

Limits	C_{obo} V_{CB} = -6 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	C_{ibo} V_{EB} = -6 V dc I_C = 0 100 kHz ≤ f ≤1 MHz
	pF	pF
Min		
Max	10	6

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Symbol	Inches		Millimeter		Note
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.065	.085	1.65	2.16	
HD	.209	.230	5.31	5.84	
LC	.707 TP		1.80 TP		4
LD		.021		.53	2, 6
LL	.500		12.70		6
LU	.016	.019	.41	.48	3, 6
L_1		.050		1.27	
L_2	.250		6.35		
TL	.028	.048	.71	1.22	
TW	.036	.046	.91	1.17	
P	.100		2.54		
Q		.040		1.02	
r		.007		.18	
α	45° TP		45° TP		

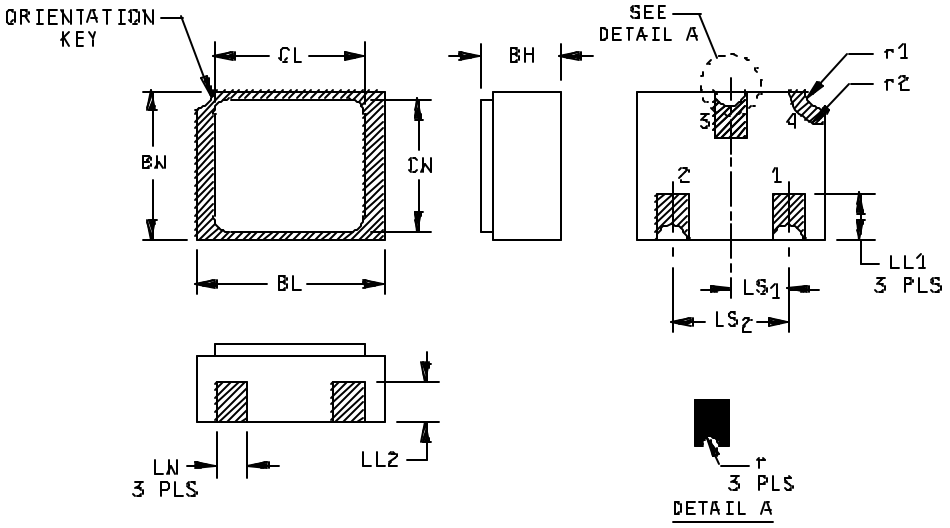


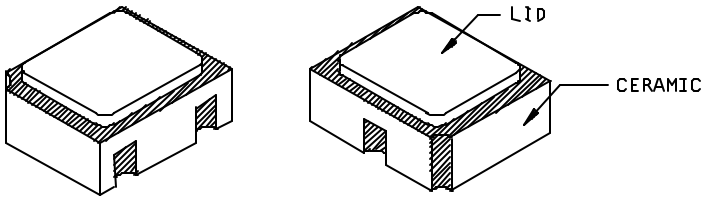
NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LU applies between L_1 and L_2. Dimensions LD applies between L_2 and LL minimum. Lead diameter shall not exceed .042 inch (1.07 mm) within L_1 and beyond LL minimum.
8. Lead designation, shall be as follows:
   1 - Emitter; 2 - Base; 3 – Collector.
9. Collector is electrically connected to case.
10. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
11. Symbol r applied to both inside corners of tab.
12. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to T0-46).

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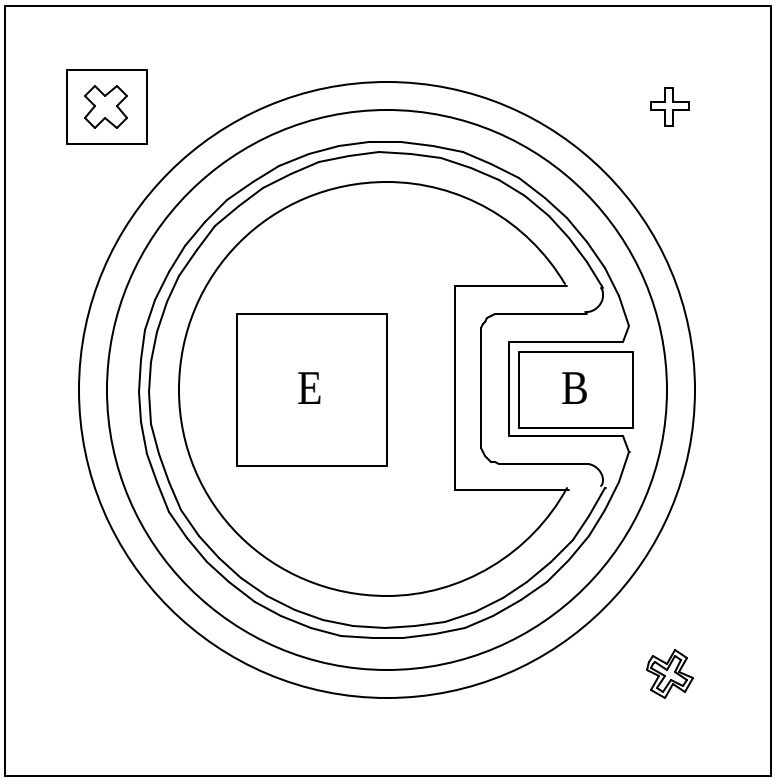
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	.046	.056	1.17	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Hatched areas on package denote metallized areas (tungsten with gold plating 60 micro inches min over 80 micro inches min nickel.
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.

* FIGURE 2. Physical dimensions for UB, surface mount.

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Die size:	.020 x .020 inch (0.508 mm x 0.508 mm).
Die thickness:	.008 ± .0016 inch (0.203 mm x 0.041 mm).
Base pad:	.002 x .003 inch (0.051 mm x 0.076 mm).
Emitter pad:	.004 x .004 inch (0.102 mm x 0.102 mm).
Back metal:	Gold, 6500 ± 1950 Ang.
Top metal:	Aluminum, 14500 ± 2500 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7500 ± 1500 Ang.

* FIGURE 3. Physical dimensions, JANKCA2N2944A through 2N2946A die (also valid for JANHCA2N2944A-2946A).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2. Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

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3.3. Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

$V_{(BR)ECO}$ Breakdown voltage, emitter to collector, with base open-circuited.

* h_{FE} (inv) Forward-current transfer ratio except that the collector and emitter shall be interchanged in the test circuit, i.e., I_E/I_B.

* M Matched pair.

I_e Emitter current (rms).

r_{ec}(on) Small-signal emitter-collector on-state resistance.

V_{EC}(ofs) Emitter to collector offset voltage, i.e., open-circuit voltage between emitter collector when the base-collector junction is forward-biased.

V_{ec} Emitter to collector voltage (rms).

* 3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (similar to T0-46), figure 2 (surface mount), and figure 3 (die).

3.4.1 Lead finish. Unless otherwise specified herein, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8. Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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* 4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the associated specification that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot to this revision to maintain qualification.

* 4.3 Screening (JANS, JAN, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JAN, JANTX and JANTXV
3c	Required, method 3131 of MIL-STD-750.	Required, method 3131 of MIL-STD-750.
* 7	Optional	Optional
9	I_{CBO1} and h_{FE} $(inv)_1$	Not applicable
11	I_{CBO1} ; h_{FE} $(inv)_1$; ΔI_{CBO1} = 100 percent of initial value or 0.2 nA dc for 2N2944 and 2N2945, 0.5 nA dc for 2N2946 Δh_{FE} $(inv)_1$ = 25 percent of initial value.	I_{CBO1} and h_{FE} $(inv)_1$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 0.2 nA dc for 2N2944 and 2N2945, 0.5 nA dc for 2N2946; Δh_{FE} $(inv)_1$ = 25 percent of initial value.	Subgroups 2 of table I herein; ΔI_{CBO1} = 100 percent of initial value or 0.2 nA dc for 2N2944A and 2N2945A, 0.5 nA dc for 2N2946A; Δh_{FE} $(inv)_1$ = 25 percent of initial value.
* 14 a and b	Required	Required

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5. T_J = 135°C minimum. 75 percent of rated power minimum.

2N2944A: V_{CB} = 10V dc.
2N2945A: V_{CB} = 10V dc.
2N2946A: V_{CB} = 10V dc.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in VIa (JANS) of 19500 and 4.4.2.1 herein. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.6 herein. See 4.4.2.2 herein for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) requirements shall be in accordance with table I, subgroup 2 herein. Delta requirements shall be in accordance with 4.5.6 herein.

* 4.4.2.1 Group B inspection (JANS), table VIa of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Condition A.
B4	1037	2,000 cycles. No heat sink or forced air cooling on the devices shall be permitted. V_{CB} = 10 V dc.
B5	1027	(NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample). V_{CB} = 10 V dc. $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). Option 1: 96 hours minimum, sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours, sample size = 45, c = 0; adjust T_A or P_D to achieve T_J = +225°C minimum.

* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV), table VIb of MIL-PRF-19500. Electrical end-points shall be in accordance with group A, subgroup 2, herein. (1)

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum, and a minimum of P_D = 75 percent of P_T as defined in 1.3. n = 45, c = 0.
2	1039	HTRB: Test condition A, 48 hours minimum. n = 45, c = 0.
3	1032	High- temperature life (non-operating), T_A = +200°C, t = 340 hours, n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and 4.4.3.1 (JANS), and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.6 herein.

* 4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E (not applicable to UB).
C6	1037	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc. Use minimum Pd = 300 mW to achieve T_J = 150°C min. No heat sink on the device shall be permitted. n = 45, c = 0.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	2036	Test condition E (not applicable to UB).
*	C5	3131	$R_{\theta JA}$ = 435°C/W see thermal impedance curves
	C6	1037	Not applicable.

* 4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANJ, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein; delta requirements only apply to subgroup C6.

(1) Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

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* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.2 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, test condition D, except that all references to the collector and the emitter of the transistor shall be interchanged.

4.5.3 Forward-current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and the emitter of the transistor shall be interchanged in the test circuit. Then: h_{FE} (inv) = I_E / I_B

4.5.4 Emitter to collector offset voltage. The transistor shall be tested in the circuit of figure 4. The base current shall be adjusted to the specified value. The voltage between the emitter and collector shall then be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.5 Small-signal emitter-collector on-state resistance. The transistor shall be tested in the circuit of figure 5. The base current shall be adjusted to the specified value and an ac sinusoidal signal current, I_e , of the specified rms value shall be applied between the emitter and collector. The rms voltage, V_{ec} , between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small-signal emitter-collector on-state resistance shall then be determined as follows:

$$r_{ec} \text{ (on)} = V_{ec} / I_e$$

where V_{ec} is the rms voltage between the emitter and collector.

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4.5.6 Delta requirements. Delta requirements shall be as specified below: (1) (2))3)

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current 2N2944A 2N2945A 2N2946A	3036	Bias condition D: I_E = 0 V_{CE} = -15 V dc V_{CE} = -25 V dc V_{CE} = -40 V dc	ΔI_{CBO1}	100 percent of initial value.		
2	Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_B = 200 µA dc; (see 4.5.3)	$\Delta h_{FE}(inv)_1$	25 percent of initial value.		

(1) The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroup 5, see 4.5.6, steps 1 and 2.

* (2) The electrical measurements for 4.4.2.2 all steps are as follows: see 4.5.6, step 2.

(3) The electrical measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, step 1 and step 2 (JANS) and step 2 (JAN, JANTX and JANTXV).

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical 3/ examination	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to 3/ 4/ 5/ solvent	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = o				
Electrical measurements 4/		Table I, subgroup 2 herein				
* Heremetic seal 4/ 6/	1071	n = 22 devices, c = 0				
Fine leak Gross leak						
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = -10 µA dc	$V_{(BR)CEO}$			
2N2944A, UB				-10		V dc
2N2945A, UB				-20		V dc
2N2946A, UB				-35		V dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
2N2944A, UB		V_{CB} = -15 V dc			10	µA dc
2N2945A, UB		V_{CB} = -25 V dc			10	µA dc
2N2946A, UB		V_{CB} = -40 V dc			10	µA dc
Emitter to base cutoff current	3061	Bias condition D	I_{EBO1}		10	µA dc
2N2944A, UB		V_{EB} = -15 V dc				
2N2945A, UB		V_{EB} = -25 V dc				
2N2946A, UB		V_{EB} = -40 V dc				

See footnotes at the end of table.

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TABLE I. Group A inspection. – Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued.						
Breakdown voltage, emitter to collector	3011	Bias condition B; I_E = -10 µA dc; I_B = 0; (see 4.5.2)	$V_{(BR)ECO}$			
2N2944A, UB				-10		V dc
2N2945A, UB				-20		V dc
2N2946A, UB				-35		V dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}			
2N2944A, UB		V_{CB} = -12 V dc			-0.1	nA dc
2N2945A, UB		V_{CB} = -20 V dc			-0.2	nA dc
2N2946A, UB		V_{CB} = -32 V dc			-0.5	nA dc
Emitter to base cutoff current	3061	Bias condition D	I_{EBO2}			
2N2944A, UB		V_{EB} = -12 V dc			-0.1	nA dc
2N2945A, UB		V_{EB} = -20 V dc			-0.2	nA dc
2N2946A, UB		V_{EB} = -32 V dc			-0.5	nA dc
Forward-current transfer ratio	3076	V_{CE} = -0.5 V dc; I_C = -1.0 mA dc	h_{FE1}			
2N2944A, UB				100		
2N2945A, UB				70		
2N2945AM, UB				70	200	
2N2946A, UB				50		
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_E = -200 µA dc (see 4.5.3)	h_{FE} (inv)$_1$			
2N2944A, UB				50		
2N2945A, UB				30		
2N2946A, UB				20		
Emitter to collector offset voltage		I_B = -200 µA dc; I_E = 0; (see 4.5.4 and figure 9).	V_{EC} (ofs)$_1$			
2N2944A, UB					-0.3	mV dc
2N2945A, UB					-0.5	mV dc
2N2946A, UB					-0.8	mV dc

See footnotes at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - continued.						
Emitter to collector offset voltage		I_B = -1 mA dc; I_E = 0; (see 4.5.4 and figure 9).	V_{EC} $(ofs)_2$			
2N2944A, UB					-0.6	mV dc
2N2945A, UB					-1.0	mV dc
2N2946A, UB					-2.0	mV dc
Emitter to collector offset voltage		I_B = -2 mA dc; I_E = 0; (see 4.5.4 and figure 9).	V_{EC} $(ofs)_3$			
2N2944A, UB					-1.0	mV dc
2N2945A, UB					-1.6	mV dc
2N2946A, UB					-2.5	mV dc
Subgroup 3						
High-temperature operation:		T_A = +100°C				
Collector to base cutoff current	3036	Bias condition D; I_C = 0	I_{CBO3}			
2N2944A, UB		V_{CB} = -15 V dc			1	nA dc
2N2945A, UB		V_{CB} = -25 V dc			2	nA dc
2N2946A, UB		V_{CB} = -40 V dc			2.5	nA dc
Emitter to base cutoff current	3061	Bias condition D; I_C = 0	I_{EBO3}			
2N2944A, UB		V_{CB} = -15 V dc			1	nA dc
2N2945A, UB		V_{CB} = -25 V dc			1.5	nA dc
2N2946A, UB		V_{CB} = -40 V dc			2.0	nA dc

See footnotes at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - continued						
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = -0.5 V dc; I_C = -1 mA dc	h_{FE2}			
2N2944A, UB				35		
2N2945A, UB				25		
2N2946A, UB				20		
Subgroup 3 - continued.						
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_E = -200 µA dc (see 4.5.3)	h_{FE} (inv)$_2$	25		
2N2944A, UB				15		
2N2945A, UB				10		
2N2946A, UB						
Subgroup 4						
Small-signal emitter-collector on-state resistance		I_B = -100 µA dc; I_E = 0; I_e = 100 µA ac (rms) f = 1 kHz (see 4.5.5 and figure 10)	r_{ec} (on)$_1$			
2N2944A, UB					10	ohm
2N2945A, UB					12	ohm
2N2946A, UB					14	ohm
Small-signal emitter-collector on-state resistance		I_B = -1 mA dc; I_E = 0; I_e = 100 µA ac (rms) f = 1 kHz (see 4.5.5 and figure 10).	r_{ec} (on)$_2$			
2N2944A, UB					4	ohm
2N2945A, UB					6	ohm
2N2946A, UB					8	ohm

See footnotes at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - continued						
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio		V_{CE} = -6 V dc; I_C = -1 mA dc; f = 1 MHz	h_{fe}			
2N2944A, UB				15	55	
2N2945A, UB				10	55	
2N2946A, UB				5	55	
Open circuit output capacitance	3236	V_{CB} = -6 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		10	pF
Input capacitance (output open-circuited)	3240	V_{EB} = -6 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz (see 4.5.1)	C_{ibo}		6.0	pF
Subgroup 4 - continued.						
Pulse response:						
Delay time	3251	Test condition B (see figure 11)	t_d		50	ns
Rise time	3251	Test condition B (see figure 11)	t_r		100	ns
Storage time	3251	Test condition B (see figure 11)	t_s		350	ns
Fall time	3251	Test condition B (see figure 11)	t_f		100	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests.
3/ Separate samples may be used.
4/ Not required for JANS.
5/ Not required for laser marked devices.
*6/ Hermetic seal test is an end-point to temperature cycling in addition to electrical measurements.

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* TABLE II. Group E inspection (all quality levels) – for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal			
Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and paragraph 4.5.6 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles.	
Electrical measurements		See table I, subgroup 2 and paragraph 4.5.6 herein.	
Subgroups 3			
Not applicable			
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their (typical) design thermal impedance curves. In addition, test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report.	
Subgroups 5, 6 and 7			Not applicable
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition A for devices ≥ 400 V Condition B for devices < 400 V	

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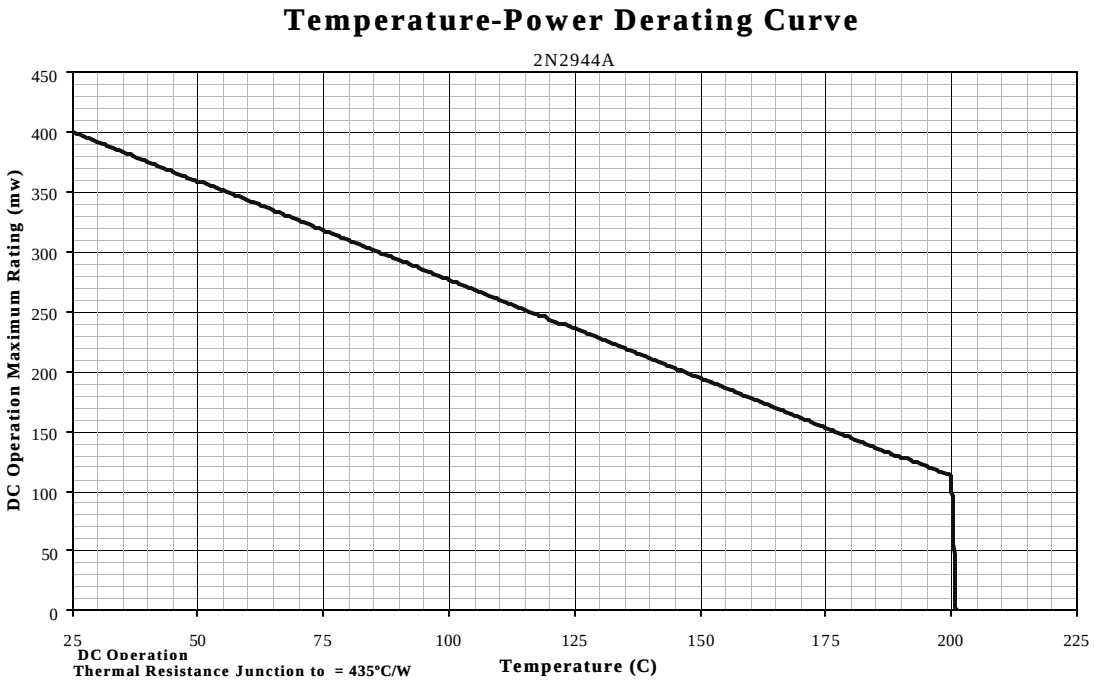


*FIGURE 4. Derating for 2N2944A ($R_{\theta JA}$), base case mounted (TO-46).

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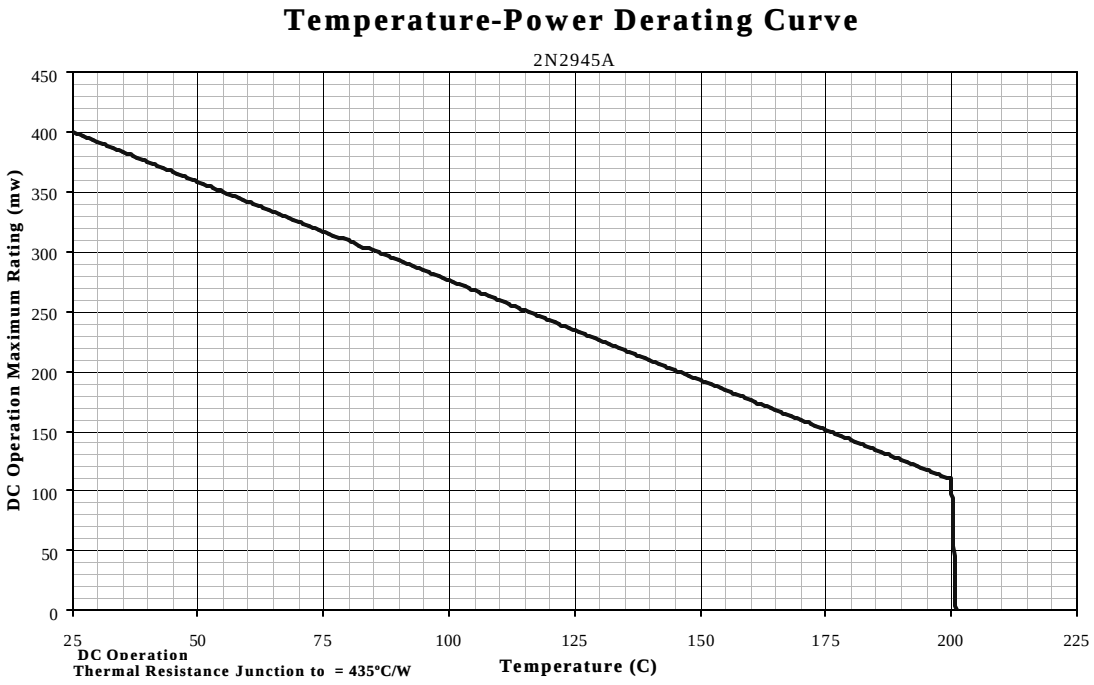


*FIGURE 5. Derating for 2N2945A ($R_{\theta JA}$), base case mounted (TO-46).

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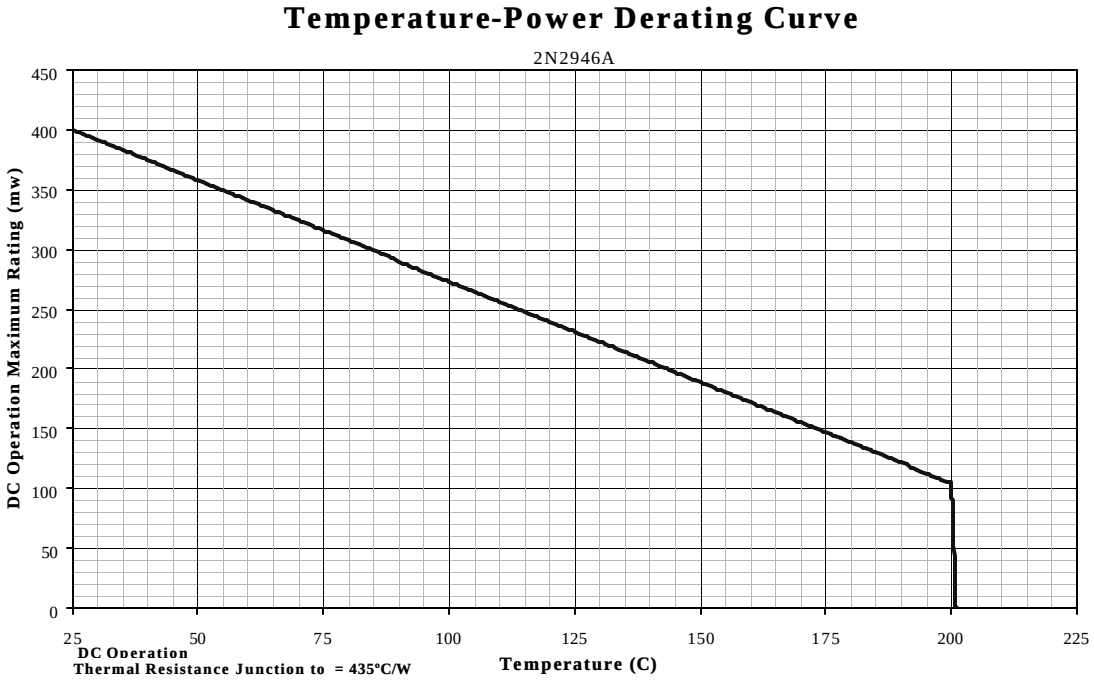
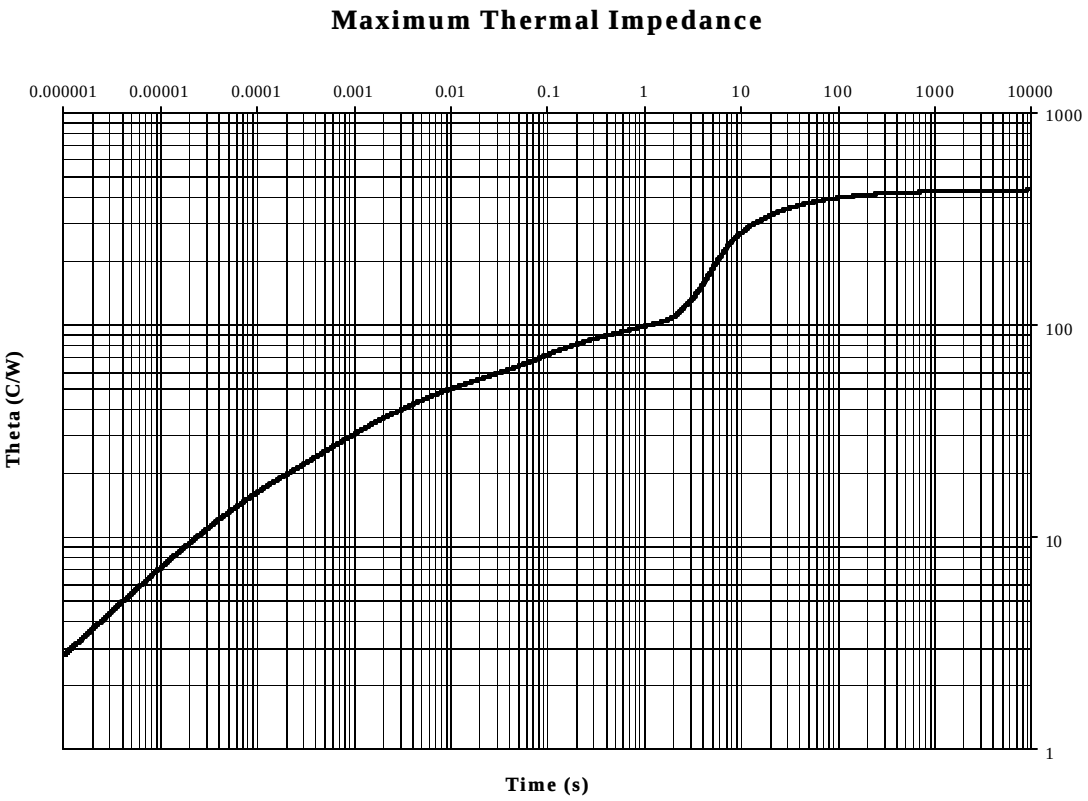


*FIGURE 6. Derating for 2N2946A ($R_{\theta JA}$), base case mounted (TO-46).

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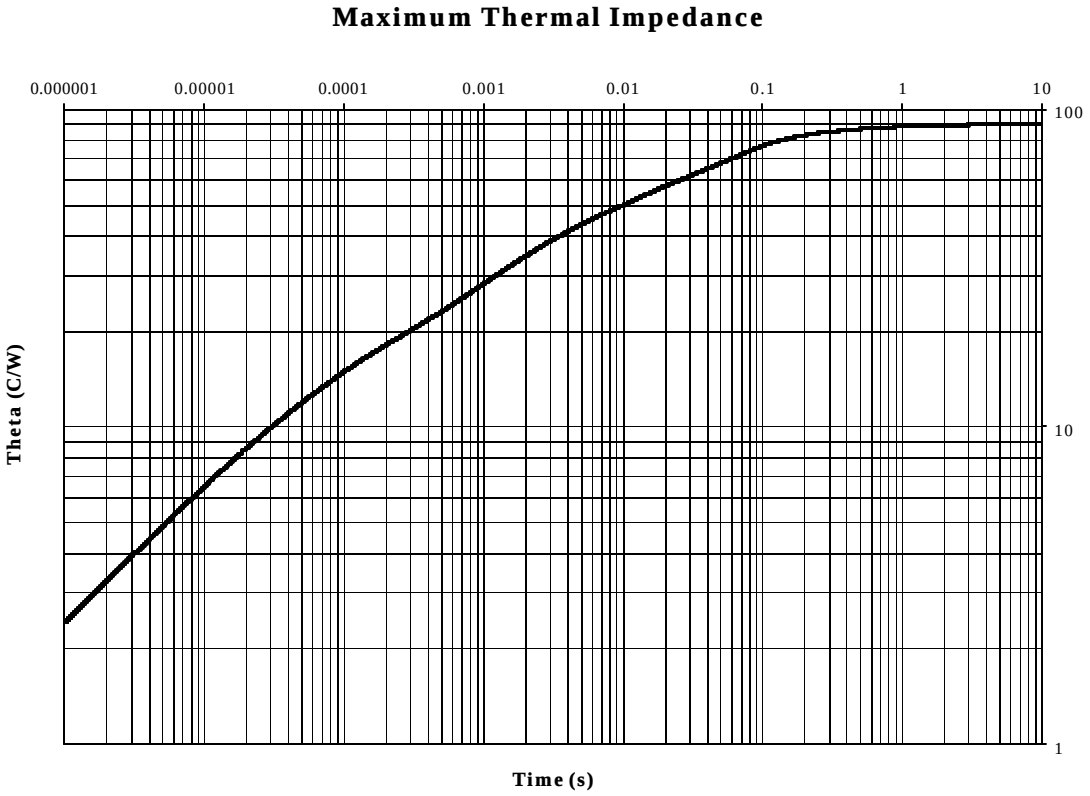


TO-46 Free Air Ta=25C with 16x23 mil chip.

Ta = 25C, Pdiss=400mW, 435C/W (Ambient thermal resistance varies with power)

*FIGURE 7. Thermal impedance graph ($R_{\theta JA}$) for (TO-46).

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Thermal resistance = 90°C/W

Solder mounted to heavy copper clad PCB at T_C =25°C.

*FIGURE 8. Thermal impedance graph (UB).

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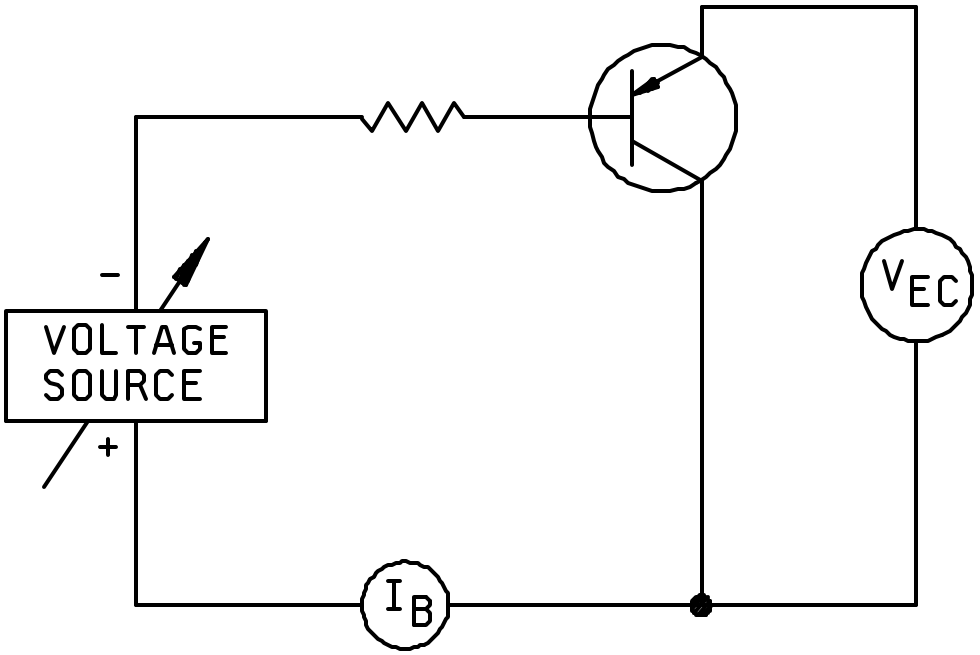


FIGURE 9. Emitter to collector offset voltage test circuit.

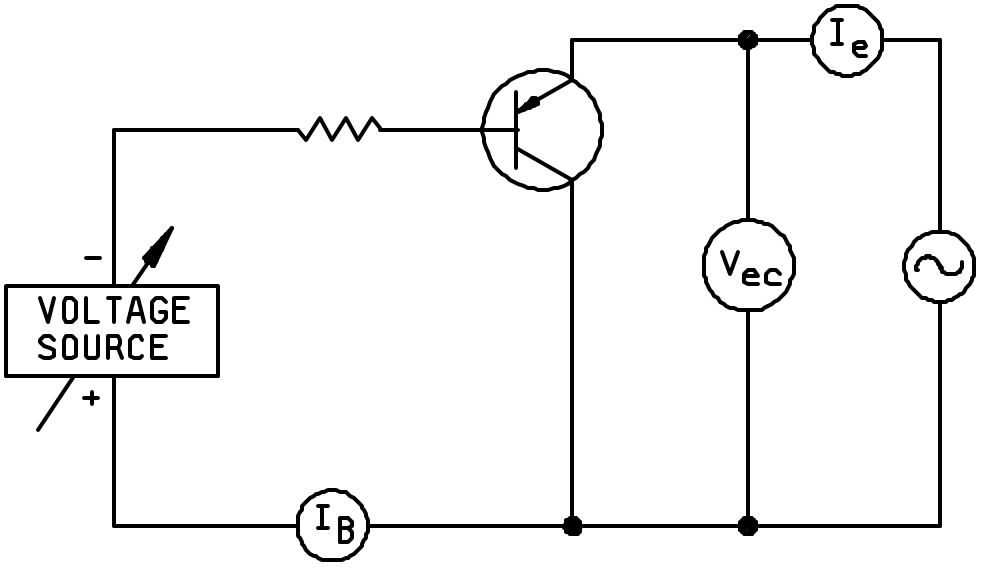
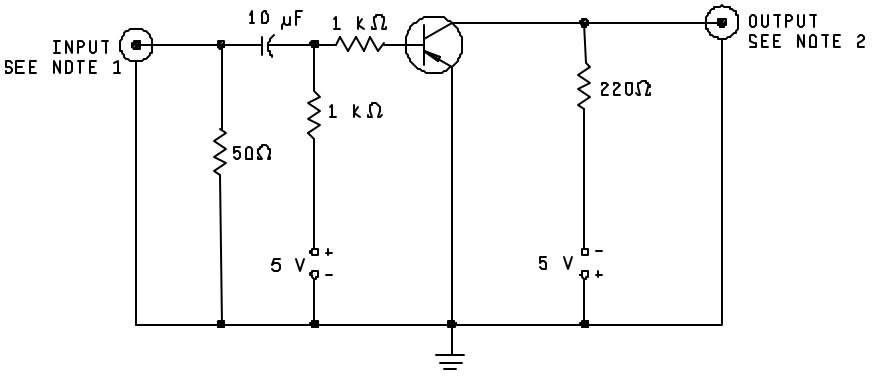
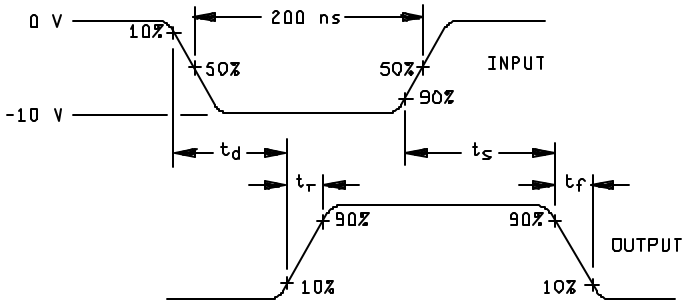


FIGURE 10. Small-signal emitter to collector on set voltage test circuit.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:

1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be ≤ 10 ns, duty cycle ≤ 2 percent. The input pulse width shall be 200 ns.
2. Output monitored with an oscilloscope with the following characteristics: Z_{in} ≤ 1 MΩ, t_r ≤ 1 ns.

FIGURE 11. Pulse response test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.2.1).

c. The lead finish as specified (see 3.4.1).

d. Type designation and quality assurance level.

e. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturer's List (QML) No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

* 6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2945) will be identified on the QML.

Die ordering information	
PIN	Manufacturer
	34156
2N2944A	JANHCA2N2944A, JANKCA2N2944A
2N2945A	JANHCA2N2945A, JANKCA2N2945A
2N2946A	JANHCA2N2946A, JANKCA2N2946A

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6.5 Changes from previous issue. The margins of this revision are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army – AV, MI
Air Force – 19, 71, 99

Preparing activity:
DLA -CC

(Project 5961-2526)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/382D	2. DOCUMENT DATE 12 May 2003

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPE 2N2944A, 2N2945A, 2N2945AM AND 2N2946A, 2N2944AUB, 2N2945AUB AND 2N2946AUB JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial DSN FAX EMAIL 614-692-0510 850-0510 614-692-6939 alan.barone@dla.mil
c. ADDRESS Defense Supply Center Columbus, ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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